RNS INSTITUTE OF TECHNOLOGY

Autonomous Institution Affiliated to VTU
RR Nagar, Bengaluru-560098

Estd : 2001

I Semester B.E. Degree Examination – Jan / Feb 2026
Introduction to AI and Applications

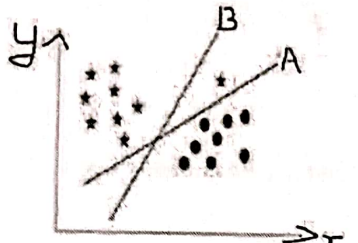
Time: 3 hrs **Max. Marks: 100**

Note:

1. ***Answer any 5 full questions, selecting at least one question from each module.***
2. ***Missing data may be suitably assumed.***
3. ***M: Marks, L: Bloom's level, C: Course Outcomes***

Q.No.		Module-1	M	L	C
Q1	a	Define the term Artificial Intelligence (AI)? Explain how it works along with two important merits and demerits.	8	L2	CO1
	b	Describe the major historical milestones in the development of AI.	6	L2	CO1
	c	What is informed search? Illustrate greedy algorithm with an example. Mention its time and space complexity.	6	L2	CO1
		OR			
Q2	a	Elaborate on the classification of AI based on functionalities and capabilities.	8	L2	CO1
	b	Define Agent and Environment with reference to AI. Explain goal-based agent along with a block diagram.	6	L2	CO1
	c	Describe the architecture and operations of a knowledge-based agent.	6	L2	CO1
		Module-2			
Q3	a	What is Prompt Engineering? Explain the different components of Prompt Engineering with an example.	8	L2	CO2
	b	Elaborate on the applications of Prompt engineering.	6	L2	CO2
	c	Build customer support using prompt engineering to guide the customer to choose a laptop. Further refine the prompt to help customers to choose with desired features, at specified range of cost.	6	L3	CO2
		OR			
Q4	a	Generate Zero, One and Few shot prompts to understand the Solar Eclipse. Analyze how the quality, detail, and accuracy of the AI's response change across the different prompting methods.	8	L3	CO2
	b	With example discuss any three text generation prompts.	6	L2	CO2
	c	Elaborate the role of effective communication in improving prompt engineering outcome.	6	L2	CO2
		Module-3			
Q5	a	Define Machine Learning. Discuss the three basic classifications of machine learning.	8	L2	CO3
	b	Explain conditional probability and the Bayes' rule.	6	L2	CO3
	c	Write a brief note on pros and cons of Neural Networks.	6	L2	CO3

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		OR			
Q6	a	Write an explanatory note on (i) Linear regression (ii) Logistic regression and (iii) Polynomial regression	8	L2	CO3
	b	Explain the working of K-means clustering algorithm.	6	L2	CO3
	c	Explain how Support vector machine works. In the Fig. Q6(c) shown, identify the boundary or hyperplane which can be used for classification  **Fig. Q6(c)**	6	L3	CO3
		Module-4			
Q7	a	Explain the concept of AI as a Service (AIaaS) along with two advantages and two challenges of using it in organizations.	6	L2	CO4
	b	List and explain any two risks associated with Artificial Intelligence emphasizing their societal impact.	4	L2	CO4
	c	Define AI Bias. Describe the sources of bias, real-world examples, and methods to mitigate the same in AI systems.	10	L2	CO4
		OR			
Q8	a	i) Explain the role and working of the Inference Engine in an Expert System. ii) Differentiate between forward and backward chaining with examples.	10	L2	CO4
	b	What is Neuromorphic Computing? Explain its architecture, features resembling the human brain, and how it contributes toward achieving Artificial General Intelligence (AGI).	10	L2	CO4
		Module-5			
Q9	a	Write the Applications of Robotics in AI.	4	L2	CO5
	b	Can AI be applied in Retail domain? Explain.	8	L2	CO5
	c	Elaborate on applying AI in Experimentation and Multi-Disciplinary Research.	8	L2	CO5
		OR			
Q10	a	Explain No Code AI and Low Code AI with an example.	8	L2	CO5
	b	Illustrate how Artificial Intelligence can be applied in education and discuss the challenges faced in its implementation.	8	L2	CO5
	c	Explain Drone using AI with an example.	4	L2	CO5

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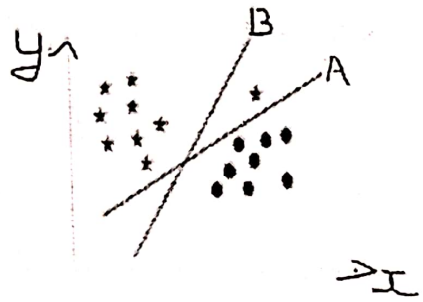
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CIE–Test II

m: I sem (Physics Cycle)
bject: Introduction to AI and its Applications
bject Code: BAIAP105

Date: 27 /12/2025
Time: 1:30pm to 3:00pm
Max.Marks:50

Instruction: Answer any 5 full questions, selecting ONE question from each part.

Q.No.		Question	Marks	RBT*	COs
1	a	Explain the following AI techniques: i) Machine Learning ii) Deep Learning and iii) Natural Language processing.	06	L2	CO3
	b	Write the diagram of neural network model and explain the prominent aspects involved in prediction/classification of outputs.	04	L2	CO3
		OR			
2	a	Explain the major components of an Expert System with their functions.	06	L2	CO4
	b	List and explain the risks associated with Artificial Intelligence and their societal	04	L2	CO4
3	a	Explain conditional probability and the Bayes rule.	06	L2	CO3
	b	Explain how Support vector machines work. In the figure shown below, identify the boundary or hyperplane which can be used for classification. 	04	L2	CO3
		OR			
4	a	Write a note on (i) Linear regression (ii) Logistic regression and (iii) Polynomial regression	06	L2	CO3
	b	Write a note on (i) Supervised Learning (ii) Reinforcement Learning	04	L2	CO3
5	a	Explain K-Nearest Neighbors algorithm. Mention its advantages and disadvantages.	10	L3	CO3

Student	Academic Score	EC Score	
A	8	7	Outstanding
B	6	5	Academically Strong
C	9	8	Outstanding
D	4	4	Average.

Classify the new student with score (7,6) assuming K=3.

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		OR			
6	a	Explain the role and working of the Inference Engine in an Expert System. Differentiate between forward and backward chaining with examples.	10	L3	CO4
7	a	What is Neuromorphic Computing? Explain its architecture, features resembling the human brain, and how it contributes toward achieving Artificial General Intelligence (AGI).	10	L5	CO4
		OR			
8	a	Analyze the differences between No-Code AI and Low-Code AI platforms, and explain their significance in accelerating AI adoption with relevant examples.	06	L3	CO5
	b	Explain Drone using AI with an example.	04	L3	CO5
9	a	Explain Applications of AI in Agriculture and its Challenges.	6	L3	CO5
	b	What is Robotics and Explain its Applications in AI	4	L3	CO5
		OR			
10	a	Explain AI in Experimentation and Multi-Disciplinary Research.	06	L2	CO5
	b	Explain Applications of AI in Education and its Challenges.	04	L3	CO5

*RevisedBloom'sTaxonomy:L1-Remember,L2-Understand,L3-Apply,L4-Analyze,L5-Evaluate,L6-Create

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CIE–Test I

Sem: I sem (Physics Cycle)
Subject: Introduction to AI and its Applications
Subject Code: BAIAP105

Date:14/11/2025
Time: 2:00pm to 3:30pm
Max.Marks:50

Instruction: Answer any 5 full questions, selecting ONE question from each part.

Q.No.		Question	Marks	RBT*	COs
1	a	Define Artificial Intelligence and explain how it works.	05	L1	CO1
	b	Discuss the advantages and disadvantages of AI in real-world applications	05	L2	CO1
		OR			
2	a	Define Search in Artificial Intelligence. Describe the role of search strategies in problem solving	10	L2	CO2
3	a	Define Knowledge Representation and explain its importance in AI.	04	L2	CO4
	b	Describe the structure of a Knowledge-Based Agent with a neat diagram..	06	L3	CO4
		OR			
4	a	Explain Agent and Environment in AI systems	05	L2	CO2
	b	Explain the different types of knowledge used in Artificial Intelligence	05	L3	CO2
5	a	Explain the types of Uninformed Search Algorithms with suitable examples. Discuss all three algorithms in detail.	10	L3	CO2
		OR			
6	a	Explain the four types of Artificial Intelligence (AI) systems with suitable examples. Classify them according to their capabilities and functionality	10	L3	CO2
7	a	Explain different Types of Prompts with suitable examples for each.	10	L3	CO2
		OR			
8	a	Define Prompt Engineering and explain its role in large language models	06	L2	CO2
	b	Explain the evolution of prompt engineering	04	L2	CO2
9	a	Explain the concept of AI as a Service (AIaaS) with suitable examples.	06	L2	CO4
	b	Identify and explain three major ethical issues associated with AI systems	04	L2	CO4
		OR			
10	a	Summarize the recent trends in Artificial Intelligence with suitable example	06	L2	CO4
	b	Explain the major challenges associated with recent trends in Artificial Intelligence	04	L2	CO4

evised Bloom's Taxonomy: L1 - Remember, L2 - Understand, L3 - Apply, L4 – Analyze, L5 - Evaluate, L6 - Create

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